

Chapter 3 - Fundamentals of Air Conditioning and Cooling Load

FUNDAMENTALS OF AIR CONDITIONING AND COOLING LOAD

3.1 Basic Air-Conditioning System

An air-conditioning system maintains the required indoor temperature and humidity by controlling the condition of air.

During cooling, heat is removed from the room and rejected to the surroundings.

Main Components

Compressor: Compresses the refrigerant and increases its pressure.

Condenser: Rejects heat from the refrigerant to the surroundings.

Expansion Device: Reduces the pressure and temperature of the refrigerant.

Evaporator: Absorbs heat from the room air and provides cooling.

3.2 Cooling Load and Heat Gain

Cooling load is the rate at which heat must be removed from a room to maintain the required indoor conditions.

Heat gain is the addition of heat to the room.

Main Sources of Heat Gain

External Sources:

- Walls
- Roof
- Windows
- Doors
- Solar radiation
- Outdoor air

Internal Sources:

- Occupants
- Lighting
- Fans

3.3 Sensible and Latent Heat

Heat gain is divided into sensible heat and latent heat.

Sensible Heat

Sensible heat causes a change in temperature without changing the moisture content.

Examples:

- Heat through walls and roof

- Solar heat

- Lighting

- Electrical equipment

Latent Heat

Latent heat is related to a change in moisture content.

Examples:

- Moisture released by occupants

- Moisture entering with outdoor air

Therefore:

$$\text{Total Heat Load} = \text{Sensible Heat Load} + \text{Latent Heat Load}$$

3.4 Major Components of Cooling Load

The important components considered in cooling-load calculation are:

1. Walls and Roof

Heat enters through walls and roof because of the temperature difference between indoor and outdoor conditions.

It depends on:

- Surface area

- Material and construction

- Temperature difference

- Insulation

2. Windows

Windows receive solar radiation and also allow heat transfer through the glass.

3. Doors and Openings

Heat can enter through doors and other openings.

Open doors can also allow outdoor air to enter the room.

4. Occupants

People release sensible and latent heat. The heat gain depends mainly on the:

- Number of occupants

- Activity level

5. Lighting and Equipment

Lights and electrical equipment release heat during operation. The load mainly depends on their:

- Number

- Power rating

- Operating condition

3.5 Ventilation, Infiltration and Factors Affecting Cooling Load

Ventilation

Ventilation is the intentional supply of outdoor air into a room.

Infiltration

Infiltration is the uncontrolled entry of outdoor air through doors, windows, gaps and

other openings.

Outdoor air can increase both sensible and latent cooling load.

Factors Affecting Cooling Load

- Room dimensions

- Wall and roof construction

- Outdoor temperature

- Solar radiation

- Doors and windows

- Number of occupants

- Lighting and equipment

- Ventilation and infiltration

3.6 Importance of Accurate Cooling Load Calculation

Accurate cooling-load calculation helps in selecting an appropriate AC capacity.

Underestimation of Load

- Insufficient cooling

- Required room conditions may not be maintained

Overestimation of Load

- Oversized AC

- Higher initial cost

- Possible inefficient operation

Therefore, actual room dimensions and operating conditions should be used wherever possible.

For this project, the HOD cabin will be surveyed to obtain:

- Room dimensions

- Doors and windows

- Openings

- Occupants

- Lighting

- Electrical equipment

The collected data will then be used for calculating the actual cooling load.

$$\text{Total Cooling Load} = \text{Sensible Load} + \text{Latent Load}$$